

Supplementary Material for: Toward Stable and Clinically Reliable AI for Whole-Slide Detection of Prostate Cancer

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Supplementary Methods

Patient-disjoint out-of-fold configuration analysis

Five patient-disjoint folds were fixed before configuration experiments. All whole-slide images (WSIs) from the same patient were assigned to one fold. For a configuration and fold k , the attention-based multiple-instance learning (AB-MIL) classifier was trained on folds other than k , model selection used only the training/validation data available within that training run, and predictions were generated for fold k . Concatenating the five held-out partitions yielded one out-of-fold (OOF) prediction per eligible development WSI. Of 2,235 development WSIs, 2,231 (1,349 patients) had complete OOF predictions across the configurations summarized in the primary analysis. OOF prediction files were required to contain unique slide identifiers, patient identifiers, fold assignments, binary reference labels, and complementary class probabilities.

For every configuration, a fixed malignancy-probability threshold of 0.5 was used. Accuracy, balanced accuracy, area under the receiver operating characteristic curve (AUROC), macro-F1, sensitivity, specificity, positive predictive value, and negative predictive value were calculated after pooling the five OOF partitions. Confidence intervals (CIs) were obtained from 2,000 patient-level bootstrap replicates, preserving all WSIs from each sampled patient. For a pair of configurations with aligned slide and patient identifiers, the same sampled patients were used for both configurations in each replicate. The paired difference was summarized by the 2.5th and 97.5th percentiles. The two-sided bootstrap P value was twice the smaller proportion of replicate differences at or below zero or at or above zero. Tests were exploratory and no multiplicity adjustment was applied.

The no-normalization OOF predictions in the stain-processing comparison were identical to the 10 $\times$ magnification experiment and were reused rather than recomputed. Patient-aligned OOF prediction provenance was retained for unnormalized, Macenko, and Reinhard comparisons; Vahadane is reported from the available aggregate OOF summary and fivefold validation output but was excluded from paired patient-bootstrap inference.

Whole-slide processing and feature extraction

Tissue regions were identified by Otsu thresholding in the CLAM preprocessing workflow. Background regions were removed, and nonoverlapping tissue fields were extracted. Slide resolution was mapped to an equivalent 20 $\times$ reference using microns-per-pixel metadata. Regions of 224 $\times$ 224, 448 $\times$ 448, or 896 $\times$ 896 pixels at that reference level represented 20 $\times$, 10 $\times$, or 5 $\times$ fields, respectively, and were resized to 224 $\times$ 224 pixels for encoder input. Pathology foundation model parameters remained frozen. Patch embeddings and coordinates were retained as the WSI feature bag. When specified, Reinhard, Macenko, or Vahadane normalization was applied before feature extraction using the same development-cohort target image for all folds and evaluation cohorts.

Processing time was measured using four NVIDIA GeForce RTX 3080 Ti graphics processing units, an Intel Xeon Silver 4210R 2.40-GHz processor, 256 GB system memory, and four-device

parallel feature extraction with a batch size of 64. Time included WSI loading, segmentation, patch preparation, optional stain normalization, and feature extraction. MIL inference was excluded because its contribution was negligible. The H-optimus-1 $20\times$ time was reused between the foundation-model and magnification experiments because preprocessing and hardware were identical.

MIL training and ensemble construction

Eleven MIL architectures were trained on identical feature bags and folds: AB-MIL, CLAM-SB, CLAM-MB, TransMIL, WIKG-MIL, Mamba2D-MIL, AEM-MIL, MICO-MIL, MSM-MIL, TDA-MIL, and GDF-MIL. Training used Adam, an initial learning rate of 2×10^{-4} , one WSI per batch, a maximum of 50 epochs, and early stopping after 30 epochs without validation improvement. Each architecture produced five patient-disjoint fold-specific models.

For architecture a and fold k , a stable interval was a maximal run of at least five epochs for which adjacent validation macro-F1 values differed by no more than 0.003. If multiple intervals were present, the last interval was retained. The checkpoint with the greatest validation macro-F1 within the retained interval was used. If no qualifying interval existed, the best checkpoint among the final five epochs was the fallback. Stability-Prioritized Ensemble (SPE) probabilities were the equal-weight mean of the 55 selected architecture-fold probabilities.

Include-All used the peak-validation checkpoint for all 55 architecture-fold pairs. Top-5 retained the five architectures with the greatest mean peak-validation macro-F1 across folds. Random retained AEM-MIL, CLAM-MB, Mamba2D-MIL, TDA-MIL, and WIKG-MIL, selected once without replacement with a fixed seed. Random and Top-5 therefore contained 25 architecture-fold members. All ensembles used equal-weight probability averaging. The architecture sets, checkpoints, and threshold were fixed before internal or external testing.

Independent testing and uncertainty intervals

The 345-WSI internal-test cohort and three external cohorts were not used for representation selection, model selection, ensemble construction, calibration, or threshold adjustment. Metrics were calculated at the WSI level, with the patient treated as the clustering unit when multiple WSIs were available. Wilson CIs were used for sensitivity and specificity. Balanced accuracy, AUROC, macro-F1, and Cohen’s kappa were assigned 95% CIs using 5,000 patient-clustered bootstrap replicates; patients were sampled with replacement, and all eligible WSIs from each sampled patient were retained together. Bootstrap sampling of the pooled external cohort was additionally stratified by institution. For strata containing only one reference class, metrics requiring both classes were not calculated. Small and highly imbalanced strata were interpreted descriptively.

The ensemble-variability analysis used 1,000 patient-clustered resamples of the fixed internal-test cohort. The trained models, architecture sets, checkpoints, and probability-aggregation rules did not change across replicates. Balanced accuracy, macro-F1, and AUROC were summarized as mean (SD), and the coefficient of variation was calculated as SD divided by the mean. This analysis quantified sensitivity to changes in test-cohort composition after the ensemble had been locked and did not estimate variability arising from repeated training, new random initializations, or checkpoint reselection.

Cancer-area-fraction analysis

Cancer area fraction was defined as the estimated percentage of evaluable tissue occupied by carcinoma on a tumor-containing WSI. Tumor-containing WSIs in the pooled external cohort were categorized as very low ($< 1\%$), low (1% to $< 5\%$), medium (5% to $< 20\%$), or high ($\geq 20\%$).

Sensitivity, false-negative rate, and malignancy-probability distributions were calculated among positive WSIs within each stratum. Specificity was calculated once using the complete tumor-free external reference set and was used as a common reference across strata. Stratum-specific balanced accuracy was calculated as the mean of this common specificity and the corresponding sensitivity. AUROC was not calculated because each cancer-area stratum contained only tumor-containing WSIs. The analysis was exploratory, retained the fixed 0.5 classification threshold, and did not include Gleason Grade Group stratification.

Matched interlaboratory analysis

One serial section from each of 60 cases was processed using the routine local H&E staining and scanning workflow of each of six laboratories. Optical-density distributions after color deconvolution characterized interlaboratory differences in staining appearance. SPE was applied independently to all 360 WSIs without laboratory-specific adaptation. Reproducibility was summarized using laboratory-specific diagnostic metrics, pairwise classification agreement, pairwise Cohen’s kappa, agreement in at least five laboratories, and the within-case SD of malignancy probabilities. This matched serial-section design retained authentic interlaboratory variation while reducing, rather than eliminating, tissue-related confounding; it was interpreted as an assessment of the complete sectioning–staining–scanning–prediction pathway rather than stain invariance alone.

Reader study and localization

The reader subset contained 100 WSIs from the external cohorts. One pathologist with seven years of diagnostic experience and one with five years of experience independently reviewed all slides using QuPath. Neither reader had previously reviewed the study slides, and both received standardized instructions before evaluation. Readers were blinded to clinical information, immunohistochemical findings, model predictions, reference labels, and each other’s assessments. Unrestricted navigation and magnification were allowed; submitted decisions could not be changed. Reading time was recorded per WSI. The comparison was descriptive and was not designed to establish equivalence, superiority, or non-inferiority between human and AI assessment.

The auxiliary YOLO11s localization model used 7,045 training and 5,180 validation patches from 16 WSIs representing 10 patients, with patient-level separation. Two pathologists who did not participate as study readers established the reference tumor regions with access to corresponding pancytokeratin/p63 immunohistochemistry. Training used stochastic gradient descent for 500 epochs, 1536×1536 -pixel inputs, mosaic augmentation, horizontal and vertical flipping, and hue–saturation–value augmentation. At inference, predictions below 0.30 confidence were removed, followed by class-agnostic nonmaximum suppression at an intersection-over-union threshold of 0.40. Patch detections were mapped to WSI coordinates. After completing the classification task, the two study readers independently delineated suspected tumor regions while remaining blinded to the reference annotations, immunohistochemical findings, and model outputs. Reader annotations and model detections were converted to WSI-level masks and compared with the reference masks using Dice coefficient and intersection over union in a patient-disjoint 25-WSI subset containing minute or morphologically subtle foci. Localization outputs did not contribute to the SPE classification probability.

Supplementary Tables

Table S1. Cohort composition by specimen type and reference class

	Development		Internal test		External-301		External-YNZL		External-FJSL	
	B	M	B	M	B	M	B	M	B	M
Patients, n	1,349		138		30		31		45	
WSIs, n	2,235		345		147		136		238	
CNB	307	437	14	35	50	14	24	26	39	72
RP	319	671	24	248	19	12	0	43	28	75
TURP	470	31	12	12	50	2	39	4	5	19

Note. B, tumor-free; CNB, core needle biopsy; M, tumor-containing; RP, radical prostatectomy; TURP, transurethral resection of the prostate; WSI, whole-slide image. Four development WSIs lacked complete OOF predictions and were omitted only from configuration inference.

Table S2. OOF performance and processing time for configuration selection

Stage	Configuration	Accuracy	Balanced accuracy (95% CI)	AUROC (95% CI)	Macro-F1 (95% CI)	Time, s/WSI
PFM	CONCH	0.9655	0.9654 (0.9514–0.9777)	0.9870 (0.9776–0.9946)	0.9655 (0.9513–0.9776)	556.94
PFM	H-optimus-1	0.9870	0.9868 (0.9792–0.9940)	0.9991 (0.9981–0.9997)	0.9870 (0.9796–0.9940)	468.78
PFM	mSTAR	0.9843	0.9842 (0.9760–0.9919)	0.9986 (0.9973–0.9996)	0.9843 (0.9763–0.9918)	391.00
PFM	UNI2	0.9825	0.9824 (0.9714–0.9916)	0.9981 (0.9959–0.9995)	0.9825 (0.9717–0.9915)	1,109.89
PFM	UNI	0.9803	0.9802 (0.9708–0.9891)	0.9984 (0.9968–0.9995)	0.9803 (0.9708–0.9891)	399.10
PFM	Virchow2	0.9736	0.9735 (0.9625–0.9839)	0.9881 (0.9798–0.9951)	0.9735 (0.9625–0.9839)	529.95
PFM	OmiCLIP	0.9619	0.9618 (0.9499–0.9731)	0.9932 (0.9893–0.9962)	0.9619 (0.9500–0.9731)	523.21
Magnification	20×	0.9870	0.9868 (0.9786–0.9940)	0.9991 (0.9981–0.9998)	0.9870 (0.9790–0.9939)	468.78
Magnification	10×	0.9767	0.9764 (0.9647–0.9873)	0.9981 (0.9963–0.9993)	0.9767 (0.9654–0.9874)	157.94
Magnification	5×	0.9695	0.9697 (0.9549–0.9821)	0.9970 (0.9948–0.9987)	0.9695 (0.9544–0.9821)	91.10
Stain processing	Reinhard	0.9861	0.9860 (0.9772–0.9936)	0.9989 (0.9978–0.9997)	0.9861 (0.9774–0.9937)	159.14
Stain processing	None	0.9767	0.9764 (0.9647–0.9870)	0.9981 (0.9963–0.9993)	0.9767 (0.9652–0.9868)	157.94
Stain processing	Macenko	0.9736	0.9732 (0.9602–0.9846)	0.9978 (0.9957–0.9992)	0.9735 (0.9607–0.9845)	336.13
Stain processing	Vahadane ^a	0.9700	0.9697 (0.9590–0.9794)	0.9971 (0.9953–0.9985)	0.9699 (0.9595–0.9794)	320.28

Note. Metrics pool 2,231 patient-disjoint OOF predictions. CIs use 2,000 patient-level bootstrap replicates. AUROC, area under the receiver operating characteristic curve; CI, confidence interval; OOF, out-of-fold; PFM, pathology foundation model; WSI, whole-slide image. ^aVahadane was excluded from paired inference because retained aligned prediction provenance was unavailable.

Table S3. Selected paired patient-bootstrap configuration comparisons

Stage	Configuration A	Configuration B	Metric	A minus B	95% CI	Two-sided <i>P</i>
PFM	H-optimus-1	mSTAR	Balanced accuracy	0.0026	−0.0011 to 0.0063	0.181
PFM	H-optimus-1	mSTAR	AUROC	0.0005	0.0001 to 0.0009	0.003
PFM	H-optimus-1	mSTAR	Macro-F1	0.0027	−0.0009 to 0.0064	0.172
Magnification	10×	20×	Balanced accuracy	−0.0104	−0.0198 to −0.0015	0.023
Magnification	10×	20×	AUROC	−0.0010	−0.0024 to −0.0001	0.014
Magnification	10×	20×	Macro-F1	−0.0103	−0.0199 to −0.0016	0.023
Magnification	10×	5×	Balanced accuracy	0.0067	−0.0022 to 0.0174	0.138
Magnification	10×	5×	AUROC	0.0011	0.0002 to 0.0023	0.014
Magnification	10×	5×	Macro-F1	0.0071	−0.0018 to 0.0182	0.123
Stain processing	Reinhard	None	Balanced accuracy	0.0096	0.0027 to 0.0174	0.003
Stain processing	Reinhard	None	AUROC	0.0008	−0.0003 to 0.0020	0.165
Stain processing	Reinhard	None	Macro-F1	0.0094	0.0026 to 0.0175	0.003
Stain processing	Reinhard	Macenko	Balanced accuracy	0.0128	0.0063 to 0.0202	< 0.001
Stain processing	Reinhard	Macenko	AUROC	0.0011	−0.0001 to 0.0025	0.067
Stain processing	Reinhard	Macenko	Macro-F1	0.0126	0.0062 to 0.0199	< 0.001

Note. Estimates use 2,000 paired patient-level bootstrap replicates. Tests were exploratory and were not multiplicity adjusted. AUROC, area under the receiver operating characteristic curve; CI, confidence interval; PFM, pathology foundation model.

Table S4. Internal-test performance of individual MIL models and ensemble strategies

Method	Accuracy	Balanced accuracy	AUROC	Macro-F1	Kappa
AB-MIL	0.9490 $\pm$ 0.0264	0.9137 $\pm$ 0.0202	0.9693 $\pm$ 0.0066	0.9036 $\pm$ 0.0382	0.8081 $\pm$ 0.0750
TransMIL	0.9252 $\pm$ 0.0145	0.8001 $\pm$ 0.1065	0.9723 $\pm$ 0.0041	0.8197 $\pm$ 0.0667	0.6448 $\pm$ 0.1274
CLAM-MB	0.9594 $\pm$ 0.0086	0.9048 $\pm$ 0.0086	0.9819 $\pm$ 0.0033	0.9164 $\pm$ 0.0141	0.8329 $\pm$ 0.0282
CLAM-SB	0.9432 $\pm$ 0.0068	0.9186 $\pm$ 0.0071	0.9775 $\pm$ 0.0042	0.8927 $\pm$ 0.0097	0.7856 $\pm$ 0.0191
WIKG-MIL	0.9467 $\pm$ 0.0075	0.9206 $\pm$ 0.0251	0.9707 $\pm$ 0.0060	0.8980 $\pm$ 0.0137	0.7963 $\pm$ 0.0273
Mamba2D-MIL	0.9467 $\pm$ 0.0062	0.8991 $\pm$ 0.0349	0.9701 $\pm$ 0.0041	0.8931 $\pm$ 0.0161	0.7864 $\pm$ 0.0321
AEM-MIL	0.9333 $\pm$ 0.0120	0.8913 $\pm$ 0.0559	0.9754 $\pm$ 0.0053	0.8703 $\pm$ 0.0268	0.7414 $\pm$ 0.0535
MICO-MIL	0.9484 $\pm$ 0.0043	0.9100 $\pm$ 0.0144	0.9625 $\pm$ 0.0153	0.8987 $\pm$ 0.0086	0.7975 $\pm$ 0.0172
MSM-MIL	0.9368 $\pm$ 0.0155	0.9016 $\pm$ 0.0458	0.9679 $\pm$ 0.0151	0.8795 $\pm$ 0.0233	0.7601 $\pm$ 0.0460
TDA-MIL	0.8986 $\pm$ 0.0472	0.8543 $\pm$ 0.0856	0.9555 $\pm$ 0.0085	0.8147 $\pm$ 0.0637	0.6365 $\pm$ 0.1195
GDF-MIL	0.9484 $\pm$ 0.0118	0.9150 $\pm$ 0.0218	0.9737 $\pm$ 0.0040	0.9002 $\pm$ 0.0179	0.8008 $\pm$ 0.0354
Random	0.9322	0.8341	0.9713	0.8443	0.6924
Top-5	0.9333	0.8730	0.9664	0.8566	0.7167
Include-All	0.9397	0.9116	0.9681	0.8863	0.7730
SPE	0.9739	0.9266	0.9813	0.9451	0.8902

Note. Individual MIL results are mean $\pm$ SD across five fold-specific models applied to the same 345-WSI cohort. Ensembles are deterministic point estimates. AUROC, area under the receiver operating characteristic curve; MIL, multiple-instance learning; SPE, Stability-Prioritized Ensemble.

Table S5. Internal-test SPE performance by specimen type

Specimen	WSIs (B/M)	Balanced accuracy	AUROC	Sensitivity	Specificity	Macro-F1	Kappa
Overall	345 (50/295)	0.9266 (0.8766–0.9700)	0.9813 (0.9586–0.9966)	0.9932 (0.9756–0.9981)	0.8600 (0.7381–0.9305)	0.9451 (0.9068–0.9762)	0.8902 (0.8137–0.9524)
CNB	49 (14/35)	0.9643 (0.8929–1.0000)	1.0000 (1.0000–1.0000)	1.0000 (0.9011–1.0000)	0.9286 (0.6853–0.9873)	0.9744 (0.9195–1.0000)	0.9489 (0.8397–1.0000)
RP	272 (24/248)	0.9147 (0.8333–0.9792)	0.9921 (0.9802–0.9997)	0.9960 (0.9775–0.9993)	0.8333 (0.6415–0.9332)	0.9394 (0.8813–0.9884)	0.8789 (0.7626–0.9767)
TURP	24 (12/12)	0.8750 (0.7083–1.0000)	0.8681 (0.6667–1.0000)	0.9167 (0.6461–0.9851)	0.8333 (0.5520–0.9530)	0.8748 (0.7078–1.0000)	0.7500 (0.4167–1.0000)

Note. Values are estimates with 95% CIs. B/M, tumor-free/tumor-containing; CNB, core needle biopsy; RP, radical prostatectomy; SPE, Stability-Prioritized Ensemble; TURP, transurethral resection of the prostate.

Table S6. External performance of individual MIL models and ensemble strategies

Method	Accuracy	Balanced accuracy	AUROC	Macro-F1	Kappa
AB-MIL	0.9488 $\pm$ 0.0087	0.9494 $\pm$ 0.0082	0.9836 $\pm$ 0.0022	0.9488 $\pm$ 0.0087	0.8978 $\pm$ 0.0172
TransMIL	0.9258 $\pm$ 0.0295	0.9249 $\pm$ 0.0309	0.9859 $\pm$ 0.0033	0.9250 $\pm$ 0.0306	0.8510 $\pm$ 0.0598
CLAM-MB	0.9592 $\pm$ 0.0065	0.9592 $\pm$ 0.0063	0.9888 $\pm$ 0.0011	0.9592 $\pm$ 0.0065	0.9184 $\pm$ 0.0130
CLAM-SB	0.9554 $\pm$ 0.0039	0.9561 $\pm$ 0.0037	0.9867 $\pm$ 0.0044	0.9554 $\pm$ 0.0039	0.9108 $\pm$ 0.0078
WIKG-MIL	0.9546 $\pm$ 0.0076	0.9552 $\pm$ 0.0073	0.9824 $\pm$ 0.0056	0.9546 $\pm$ 0.0076	0.9093 $\pm$ 0.0151
Mamba2D-MIL	0.9500 $\pm$ 0.0089	0.9503 $\pm$ 0.0089	0.9829 $\pm$ 0.0041	0.9500 $\pm$ 0.0089	0.9000 $\pm$ 0.0177
AEM-MIL	0.9238 $\pm$ 0.0143	0.9248 $\pm$ 0.0144	0.9756 $\pm$ 0.0057	0.9238 $\pm$ 0.0144	0.8479 $\pm$ 0.0286
MICO-MIL	0.9581 $\pm$ 0.0106	0.9584 $\pm$ 0.0103	0.9786 $\pm$ 0.0059	0.9581 $\pm$ 0.0106	0.9162 $\pm$ 0.0212
MSM-MIL	0.9573 $\pm$ 0.0087	0.9577 $\pm$ 0.0083	0.9889 $\pm$ 0.0029	0.9573 $\pm$ 0.0087	0.9146 $\pm$ 0.0174
TDA-MIL	0.9215 $\pm$ 0.0424	0.9218 $\pm$ 0.0428	0.9863 $\pm$ 0.0047	0.9207 $\pm$ 0.0434	0.8430 $\pm$ 0.0850
GDF-MIL	0.9565 $\pm$ 0.0036	0.9568 $\pm$ 0.0033	0.9854 $\pm$ 0.0032	0.9565 $\pm$ 0.0036	0.9131 $\pm$ 0.0071
Random	0.8654	0.8627	0.9642	0.8618	0.7287
Top-5	0.8980	0.8962	0.9431	0.8964	0.7946
Include-All	0.9198	0.9178	0.9814	0.9187	0.8386
SPE	0.9654	0.9655	0.9912	0.9654	0.9309

Note. Individual MIL results are mean $\pm$ SD across five fold-specific models applied to the fixed 521-WSI external cohort. Ensembles are deterministic point estimates. AUROC, area under the receiver operating characteristic curve; MIL, multiple-instance learning; SPE, Stability-Prioritized Ensemble.

Table S7. External SPE performance by institution and specimen type

Grouping	Stratum	WSIs (B/M)	Balanced accuracy	Sensitivity	Specificity	AUROC	Macro-F1	Kappa
Institution	301	147 (119/28)	0.9202 (0.8529–0.9779)	0.8571 (0.6851–0.9430)	0.9832 (0.9408–0.9954)	0.9925 (0.9811–0.9991)	0.9319 (0.8795–0.9779)	0.8639 (0.7597–0.9559)
Institution	YNZL	136 (63/73)	0.9773 (0.9477–1.0000)	0.9863 (0.9264–0.9976)	0.9683 (0.8914–0.9913)	0.9941 (0.9811–1.0000)	0.9778 (0.9482–1.0000)	0.9556 (0.8964–1.0000)
Institution	FJSL	238 (72/166)	0.9572 (0.9285–0.9840)	0.9699 (0.9314–0.9871)	0.9444 (0.8657–0.9782)	0.9887 (0.9767–0.9982)	0.9554 (0.9262–0.9802)	0.9107 (0.8524–0.9605)
Specimen	CNB	225 (113/112)	0.9644 (0.9378–0.9867)	0.9554 (0.8997–0.9808)	0.9735 (0.9248–0.9909)	0.9913 (0.9787–0.9994)	0.9644 (0.9378–0.9867)	0.9289 (0.8756–0.9733)
Specimen	RP	177 (47/130)	0.9498 (0.9062–0.9855)	0.9846 (0.9456–0.9958)	0.9149 (0.8007–0.9664)	0.9943 (0.9859–0.9997)	0.9559 (0.9162–0.9855)	0.9119 (0.8327–0.9710)
Specimen	TURP	119 (94/25)	0.9347 (0.8692–0.9947)	0.8800 (0.7004–0.9583)	0.9894 (0.9422–0.9981)	0.9685 (0.9047–1.0000)	0.9478 (0.8922–0.9875)	0.8957 (0.7848–0.9750)
Overall	Pooled external	521 (254/267)	0.9655 (0.9483–0.9809)	0.9625 (0.9324–0.9795)	0.9685 (0.9391–0.9840)	0.9912 (0.9826–0.9979)	0.9654 (0.9482–0.9808)	0.9309 (0.8963–0.9616)

Note. Values are estimates with 95% CIs. B/M, tumor-free/tumor-containing; CNB, core needle biopsy; RP, radical prostatectomy; SPE, Stability-Prioritized Ensemble; TURP, transurethral resection of the prostate.

Table S8. External-test performance by cancer area fraction

Outcome	Very low (< 1%)	Low (1%–< 5%)	Medium (5%–< 20%)	High ($\geq$ 20%)
Sensitivity	88.14%	97.44%	100.00%	97.44%
False-negative rate	11.86%	2.56%	0.00%	2.56%
Specificity ^a	96.85%	96.85%	96.85%	96.85%
Balanced accuracy ^a	92.50%	97.15%	98.43%	97.15%
Median malignancy probability (IQR)	0.9748 (0.7902–0.9962)	0.9974 (0.9922–0.9989)	0.9991 (0.9972–0.9995)	0.9987 (0.9941–0.9997)

Note. Cancer area fraction is the estimated percentage of evaluable tissue occupied by carcinoma. Sensitivity, false-negative rate, and malignancy-probability summaries were calculated among tumor-containing WSIs within each stratum. ^aSpecificity was calculated using the complete set of 254 tumor-free WSIs in the pooled external cohort and was used as a common reference across strata; balanced accuracy is the mean of this common specificity and stratum-specific sensitivity. The analysis was exploratory. IQR, interquartile range.

Table S9. Diagnostic performance and case-level concordance across six laboratory workflows

Laboratory	n	Accuracy	Balanced accuracy	AUROC	Macro-F1	Kappa
301	60	0.9167	0.9167	0.9844	0.9165	0.8333
900	60	0.9167	0.9167	0.9878	0.9166	0.8333
YNZL	60	0.9333	0.9333	0.9811	0.9333	0.8667
NJJZ	60	0.9667	0.9667	0.9978	0.9667	0.9333
GXRK	60	0.9333	0.9333	0.9878	0.9333	0.8667
HNAY	60	0.9667	0.9667	0.9878	0.9666	0.9333
Mean $\pm$ SD	–	0.9389 $\pm$ 0.0228	0.9389 $\pm$ 0.0228	0.9878 $\pm$ 0.0056	0.9388 $\pm$ 0.0228	0.8778 $\pm$ 0.0455

Note. Each laboratory processed one serial section from each of 60 cases (30 tumor-free and 30 tumor-containing). Accuracy equals balanced accuracy because each laboratory-specific dataset was class balanced. Serial sections were matched but were not microscopically identical. AUROC, area under the receiver operating characteristic curve; SD, standard deviation; SPE, Stability-Prioritized Ensemble.

Table S10. Reader-study subset composition

Category	Tumor-free, n	Tumor-containing, n	Total, n
Institution			
301	17	18	35
YNZL	6	27	33
FJSL	7	25	32
Overall	30	70	100
Specimen type			
CNB	14	23	37
RP	7	36	43
TURP	9	11	20
Overall	30	70	100

Note. CNB, core needle biopsy; RP, radical prostatectomy; TURP, transurethral resection of the prostate.

Table S11. Diagnostic performance in the retrospective reader study

Reader or model	Accuracy	Balanced accuracy	Sensitivity	Specificity	Macro-F1	Kappa	AUROC	Time, s/WSI
Pathologist, 7 years	0.9500	0.9262	0.9857	0.8667	0.9387	0.8775	–	76
Pathologist, 5 years	0.7100	0.7929	0.5857	1.0000	0.7064	0.4590	–	110
SPE	0.9300	0.9214	0.9429	0.9000	0.9174	0.8349	0.9743	156.67

Note. AUROC was unavailable for readers because continuous confidence scores were not recorded. Reader times represent slide-interpretation time, whereas SPE time represents computational preprocessing and feature extraction; these quantities were not intended for direct statistical comparison. The reader comparison was descriptive and was not designed to establish equivalence, superiority, or non-inferiority. AUROC, area under the receiver operating characteristic curve; SPE, Stability-Prioritized Ensemble; WSI, whole-slide image.

Table S12. Descriptive region-level localization in 25 WSIs with minute or subtle cancer foci

Annotator or model	Dice coefficient	Intersection over union	Time, s/WSI
YOLO11s localization module	0.7735	0.7477	96.91
Pathologist, 7 years	0.7070	0.6079	—
Pathologist, 5 years	0.5321	0.4774	—

Note. The subset was patient-disjoint from localization-model training and validation and was enriched for minute or morphologically subtle cancer foci. Reference regions were established by two pathologists who did not participate as study readers and who had access to matched immunohistochemistry. Human annotation time was not standardized, and no formal human–AI superiority analysis was prespecified. WSI, whole-slide image.